

Chaos Engineering

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What and Why Chaos Engineering?

- In IT, it(no puns intended) began at *Netflix*.
- *Murphy's law*.
- Builds confidence in overall distributed systems to withstand turbulent & unexpected conditions.
- Highlights weakness of the complex system *proactively*.
- Minimal downtime -> Less *SLA* breaches -> Less *revenue loss*.
- Improves the resilience of the system. Key areas:
 - Infrastructure Failures
 - Network Failures
 - Application Failures

How Chaos Engineering differs from Testing?

- In testing, *assertions* are made.
- Assertions are typically binary, whether property is correct or not.
- Testing breaks the system in *preconceived* way.
- Chaos Engineering doesn't test known properties, it tests hypothesis.
- Chaos Engineering generates new knowledge.
 - Examples:
 - Simulating failure of entire AZ, region, datacenter.
 - Injecting latencies between services.
 - Forcing system clocks out of sync.

Designing Chaos Experiments

- Identify the steady state of the system.
- Pick a hypothesis.
- Choose the scope.
- Identify the operational metrics.
- Notify concerned members.
- Run the experiment.
- Analyze the results.
- Increase the scope.
- Automate.

What is Chaos Lambda?

- Open sourced by BBC.
- *EC2* instances are volatile(99.99% SLA).
- *AWS recommends* to place *EC2* instance under *Autoscaling* groups.
- Chaos Lambda simulates the failure of *EC2* instance in *Autoscaling* group(s).

How it works?

- **Schedule**

- **Default Value:** cron(0 10-16 ? * MON-FRI *)
- **Possible Values:** cron(0 10-16 ? * MON-FRI *)

- **Default Probability**

- **Default Value:** 0.166
- **Possible Values:** 0.0 - 1.0

- **Regions**

- **Default Value:** Current Region
- **Possible Values:** List of Regions

Demo

Thank You & QA

Reference: <https://github.com/dastergon/awesome-chaos-engineering>